Akanksha Singh

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About Me

Researcher in machine learning and deep learning with expertise in vision and speech. My broad goal is advancing responsible AI research, with a specific focus on fairness, inclusion, explainability, societal and environmental impact.

Education

Indian Institute of Science Education and Research, Bhopal — *Aug 2019 - July 2024*

BS-MS in Electrical Engineering and Computer Science | CPI : 8.36/10

- Minor: Data Science and Engineering
- Relevant Courses: Computer Vision, Deep Learning, Intelligent Robotics, Machine Learning, Natural Language Processing, Network Science

Experience

Centre for Responsible AI (CeRAI), IIT Madras — *Aug 2024 - Current*

Post-Baccalaureate Fellow | Supervisor: Prof. Balaraman Ravindran

- Leading the research of grounding artifacts of audio deepfakes into linguistic features like phonology, morphology, syntax, and semantics to devise an explainable detection model by design, tailored to the Indian context, aimed at supporting fact-checkers, media organisations, and public-interest platforms.
- Chapter author of the "Grounding Responsible AI in Agriculture", a project in collaboration with Meta OpenLoop. The work explores the current applications and associated risks of AI in agriculture, identifies key gaps in its development and deployment, and examines relevant policies, regulations, and legal frameworks. The paper emphasised the importance of responsible AI through actionable technical and policy recommendations, grounded in expert interviews and comprehensive research.
- Ongoing discussions on using a multi-agent system for multimodal deepfake detection in tandem with the RAG framework to provide additional support by fact-checking.

Image Analysis and Biometrics Lab, Department of CSE, IIT Jodhpur — *May 2023 - Apr 2024*

External Master's Thesis | Supervisor: Prof. Mayank Vatsa & Prof. Richa Singh

- Created and curated a large (approx 1.3M samples), diverse (26 generation techniques across 4 sets of real-fake and audio-video), balanced (wrt sex and skin-tone) benchmark multimodal multilingual deepfake dataset comprising identity-aware swaps and synthetic media.
- Experimented with approx 20 visual and audio deepfake models, optimised for mass generation over approx 6 months using A100, V100 and A40 GPU configurations.
- Performed qualitative experiments such as BRISQUE score for visual quality and FAD for audio quality. And benchmarked using approx 12 existing state-of-the-art unimodal (visual and audio) and multimodal deepfake detection models.

Publications

Conference
ILLUSION: Unveiling Truth with a Comprehensive Multi-Modal, Multi-Lingual Deepfake Dataset
Kartik Thakral*, Rishabh Ranjan*, **Akanksha Singh**, Akshat Jain, Mayank Vatsa, Richa Singh
International Conference on Learning Representations (ICLR) 2025
* = equal contribution

Skills

Languages Python, C/C++
Frameworks PyTorch, HuggingFace, Dockers, LaTeX, Git, Bash, Slurm

Achievements

2024 Women Post Baccalaureate Fellow, Research Fellowship, CeRAI, IIT Madras
2023 Qualified for Assistant Professor, Subject: CSA, UCG-NET

Presentations

Poster Explainable Audio Deepfake Detection, Conclave on AI Governance, CeRAI IIT Madras
Presentation Explainable Multilingual Audio Deepfake Detection, WSAI Bi-weekly, WSAI IIT Madras
Poster IILUSION, ICLR 2025
Presentation Deepfakes & ILLUSION, Group Weekly, IIT Madras

Academic Projects

Interpretable Deep Learning for Text Classification

- Implemented an existing novel interpretability framework for text CNNs from scratch to linguistically motivated features. The proposed features are borrowed from existing work on text classification of shorter texts between fiction and non-fiction genres (Brown Corpus).
- The exit criteria are to corroborate our interpretability results with the baseline results obtained from the existing work using logistic regression.

Consumer Complaints: Text Classification Problem

- Used machine learning and deep learning models for a multi-class nominal label classification.
- A vectorisation and feature selection pipeline was used for statistical architecture, and hyperparameters were fed to a cross-validation grid search on 5 supervised learning models. Logistic regression gave the best results, accuracy of 85.9%.
- For DL architecture, FFN, bi-directional LSTM, RNN, and transformers were used. Bi-directional LSTM gave the best results, with an accuracy of 86.5%.

Teaching Experience

ECS308: Data Science and Machine Learning *Jan 2023 - April 2023*
Undergraduate Teaching Assistant

- Course Instructor: Dr. Tanmay Basu. One of the two TAs for the course with 50 students.
- Conducted labs on supervised and unsupervised machine learning and hyperparameter tuning.
- Graded assignments, course projects, quizzes and examinations.

Personal Interests

Sports	Served as the first female Secretary for the Sports Society. Participated in the Inter IISER Sports Meet in the sports basketball, football, and athletics, and won Silver (2019) and Gold (2022 and 2023) in Basketball. Served as Basketball Coordinator.
Trekking	Completed high-altitude (upto 15000ft) treks.
Quiz	Served as the Publicity and PR Coordinator for the IBQC-Quizzing Society.
Volunteer	Volunteered as a Peer (Student) Counsellor to the Institute Counselling Cell.
Dance	Served as the Committee Member of the Mayuraa-Dance Society.
Arts	Served as the Vice-Secretary of the FALC-Fine Arts and Literary Society.